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Radiolucent surgical retractor

Abstract

A self-supporting surgical retractor is provided for retracting soft tissue and anatomy at a spinal surgical site. The retractor includes a tubular base defining an upper rim and a lower rim and a working channel there between the upper and lower rims, in which the base and lower rim are sized and configured to be seated on adjacent vertebral bodies spanning an intervertebral space. At least two elongated legs are provided, each projecting from the upper rim and each including a fixation feature at a free end thereof. Each of the legs has a length sufficient for the fixation feature to be outside the body of a patient when the lower rim is seated on adjacent vertebral bodies of the patient. The retractor is formed of a radio-transparent or radio-lucent material.

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Background/Summary

REFERENCE TO RELATED APPLICATION AND PRIORITY CLAIM (1) This application is a continuation of U.S. application Ser. No. 17/024,820, filed on Sep. 18, 2020, which issued as U.S. Pat. No. 11,844,504 on Dec. 19, 2023, which application is a utility filing from and claims priority to U.S. Provisional Application No. 62/903,255, filed on Sep. 20, 2019, the entire disclosure of which is expressly incorporated herein by reference.

BACKGROUND

(1) The present disclosure relates to tissue retractors for use in surgical procedures to retract tissue and/or organs at a surgical site.

(2) In general terms, a retractor is a surgical instrument used to separate the edges of a surgical incision or wound, as shown in FIG. 1. Retractors can also be configured to hold back underlying organs and tissue so that other parts of the body may be accessed by the surgeon. A wide variety of surgical retractors are available, depending on the requirements for the particular surgical procedure and, in some cases, surgeon preference. The simplest retractor is a hand-held device, such as the retractor shown in FIG. 2. The hand-held device includes a handle, an elongated body and a curved blade configured to engage the edges of the wound. This type of retractor is held by a surgeon or medical assistant throughout the surgical procedure, which can be a detriment due to the potential for fatigue. However, the hand-held retractor provides a degree of flexibility not easily accomplished by other retractors in that the tissue/organs being retracted can be directed by the operating surgeon.

(3) A variant from the hand-held retractor is the self-retaining retractor, such as the retractors shown in FIG. 3. The self-retaining retractors are typically in a scissors or sliding mount form. In these types of devices blades are provided at the ends of at least two arms, with the arms coupled to each other (scissors-type) or mounted on a separate beam (sliding mount). The self-retaining retractors are held in position at the surgical site by the tension applied to the edges of the incision and typically rest on the patient's body.

(4) A common type of retractor is the frame-type retractor, as depicted in FIG. 4. In this type of device, a rigid frame is formed to define the eventual perimeter of the surgical incision. The rigid frame is anchored to the operating table to hold the frame in the desired position. Individual retraction arms are mounted to the frame and manipulated to engage and retract the edges of the incision. A variation of the frame-type retractor is the cable-winch retractor in which the rigid frame is offset from the patient, acting as a rigid support for a cable-winch system used to pull retractor blades toward the frame. This type of frame retractor has the advantage of keeping the surgical site relatively free of equipment that can hamper access by the surgeon.

(5) Tubular retractors are another common surgical device. As depicted in FIG. 5, the tubular retractor is essentially a tube that is manipulated into the surgical site, sometimes with the assistance of peripheral retractors. The tubular retractor creates a working channel through which various devices, instruments and implants are introduced. The tubular retractors can be self-retaining or anchored to a separate rigid frame. A split-blade retractor, as shown in FIG. 6, is a variation of the tubular retractor in that the walls of the tube are outwardly movable to increase the volume of the working space defined by the retractor. The split-blade retractor starts out as a closed tube for introduction into the incision, and then expanded as needed.

(6) The typical surgical retractor is formed of a metal, such as stainless steel, in order to provide sufficient strength and rigidity, as well as sterilizability. One drawback of the conventional metallic retractor is that the metal is generally radio-opaque. In some open surgical procedures, such as spinal surgeries, it can be necessary to obtain an X-ray image in order to verify the local anatomy and/or the orientation of instruments and implants. The radio-opaque retractor component can thus obscure or even hide parts of the surgical field that may be important to the surgeon. Some retractors are made of a less radio-dense metal, such as aluminum, to retain the benefits of stainless steel with less impact on the ability to image the surgical site.

(7) The operating room is equipped with overhead lights that can be positioned and oriented to illuminate the surgical field. However, it is difficult to position the limited number of lights (usually two per OR) to avoid shadows from the surgeon(s) and attending medical personnel. One option has been to provide the surgeon with a headlight system which avoids the problem with shadows. However, it is still often difficult for the surgeon to direct the light beam(s) where it is

needed while also performing the surgical procedure. To address the problems with these approaches, some retractor systems include lighting components associated with the retractor structure, such as the system shown in FIG. 7. Fixed retractor systems can include one or more lights mounted to the same fixed frame on which the retractors are mounted. In another approach, the retractor blade itself is provided with a lighting component integrated into the blade to directly illuminate the space adjacent the retractor. One detriment is that the addition of lighting to the retractor system takes up space in the surgical site and adds complexity to the retractor set up.

(8) One typical spinal surgical procedure relies on an anterior approach in which the spine is accessed through the abdomen of the patient, as shown in FIG. 8. The size of the wound required for an anterior approach can vary, but is usually limited to a working channel diameter of 7-10 cm. Frame retractors are typically used to retract the edges of the incision, as well as to hold critical blood vessels. In particular, the aorta and vena cava must be moved and held in place to provide clear access to the lower lumbar and sacral vertebrae. Due to the nature of the surgery, the retractors may get contacted or the spine may move as it is being manipulated, which in turn may cause the blood vessels to become dislodged from the retractor and creep into the surgical field.

(9) Anterior spinal surgery also involves other specialized instrumentation, such as retractor shims, as shown in FIG. 9, and paddles, as shown in FIG. 10. Retractor shims are locked to the existing retractor blades and used to either anchor the retractor blade or increases the effective area of the blade. In the former case, the shim can include a spike or screw that can be driven into a vertebral body to hold the retractor in position during the surgery. In the latter case, the shim can be used to lengthen or widen the retractor blade to increase the amount of tissue being held back by the retractor. Shims are commonly used on split blade retractors.

(10) The paddle (FIG. 10) is an instrument that is introduced into the space between vertebral bodies after the intervertebral disc has been removed. The paddle can be a distractor that is used to separate the adjacent vertebral bodies, or a sizer that is used to determine the size of an implant to be introduced into the space. Although the paddle instrument is not part of the distractor system, it is a crucial instrument used in anterior spinal surgeries that must be accommodated by the retractor system.

(11) In view of the benefits and detriments of prior retractor systems, there remains a need for a retractor that is radio-transparent or radio-lucent yet still sturdy enough to keep the surgical space open throughout the surgery. There is also a need for a single use, disposable retractor system that eliminates the need for sterilization for re-use and reduces the overall cost of the retractor instrumentation.

SUMMARY OF THE DISCLOSURE

(12) The present disclosure provides a retractor system particularly suited for use during anterior spinal surgeries. In one feature the primary components of the retractor system are formed of a radio-transparent or radio-lucent material, such as a plastic material. At the core of the retractor system is a retractor component that has a structure similar to a tubular retractor, with additional features that help hold the retractor to the spine during the procedure and prevent the adjacent blood vessels from creeping back into the surgical field. The system includes connector components for lengthening or shortening the retractor, and shims that can be used as a disc space distractor.

Description

DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is pictorial view of one type of surgical retractor of the prior art shown in use.
- (2) FIG. 2 is a pictorial view of a hand-held surgical retractor of the prior art.
- (3) FIG. 3 are pictorial views of self-retaining surgical retractors of the prior art.

- (4) FIG. 4 is a pictorial view of a frame-type retractor system of the prior art.
- (5) FIG. 5 is a pictorial view of a tubular retractor of the prior art.
- (6) FIG. 6 is a pictorial view of a split blade retractor of the prior art.
- (7) FIG. 7 is a pictorial view of a retractor-based lighting system of the prior art.
- (8) FIG. 8 is a pictorial representation of an anterior retractor system of the prior art for use in anterior spinal surgery.
- (9) FIG. 9 is a diagram of a shim of the prior art used with surgical retractors.
- (10) FIG. 10 is a diagram of a distractor of the prior art used in spinal surgery.
- (11) FIGS. 11A-C are pictorial views of a surgical retractor system according to one embodiment of the present disclosure, showing the system relative to the patient's anatomy during an anterior spinal surgery.
- (12) FIGS. 12A-B are end and side views of a retractor for use in the retractor system shown in FIGS. 11A-C.
- (13) FIGS. 13A-D are perspective, bottom, side and end views of a retractor, according to another embodiment of the disclosure, for use in the retractor system shown in FIGS. 11A-C.
- (14) FIG. 14 is a perspective view of an engagement portion of the brackets shown in FIG. 11A.
- (15) FIGS. 15A-C are views of a retractor for use in the retractor system shown in FIGS. 20A-C.
- (16) FIG. 16 is a side view of a retractor of the present disclosure engaged to a hand-held retractor.
- (17) FIG. 17 is pictorial view of the retractor of FIGS. 13A-D in position during an anterior spinal surgery, and including a shim according to another feature of the present disclosure.
- (18) FIGS. 18A-E are perspective, side, edge and end views of the shim shown in FIG. 17.
- (19) FIG. 19 is a pictorial representation of a self-supporting retractor system according to one embodiment of the present disclosure.
- (20) FIG. 20 is an enlarged view of components of the self-supporting retractor system shown in FIG. 19.
- (21) FIGS. 21A-D are perspective, bottom, side and end views of a barrel mechanism of the self-supporting retractor system shown in FIG. 19
- (22) FIGS. 22A-D are perspective, bottom, side and end views of a strap of the self-supporting retractor system shown in FIG. 19.
- (23) FIGS. 23A-D are perspective, bottom, side and end views of a body of the barrel mechanism shown in FIG. 19A-D.
- (24) FIG. 24 is a perspective view of the body shown in FIGS. 23A-D with the strap of FIGS. 22A-D disposed therein.
- (25) FIGS. 25A-D are perspective, bottom, side and end views of a barrel of the barrel mechanism shown in FIG. 21A-D.
- (26) FIG. 26A-D are perspective, bottom, side and end views of an end cap of the barrel mechanism shown in FIG. 21A-D.
- (27) FIG. 27 is a perspective view of a self-supporting retractor system according to another embodiment of the present disclosure.
- (28) FIG. 28A, B is a perspective view of diffuse lighting systems used with the retractors disclosed herein.
- (29) FIG. 29 is a perspective view of a direct lighting system used with the retractors disclosed herein.
- (30) FIGS. 30A, B are side views of a retractor system according to a further embodiment incorporating movable legs.

DETAILED DESCRIPTION

(31) For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to the embodiments illustrated in the drawings and described in the following written specification. It is understood that no limitation to the scope of the disclosure is thereby intended. It is further understood that the present disclosure includes any alterations and

modifications to the illustrated embodiments and includes further applications of the principles disclosed herein as would normally occur to one skilled in the art to which this disclosure pertains. (32) A retractor system **10** according to the present disclosure is shown in FIGS. **11A-11C**, in which the system is a frame-based system. The system **10** is shown in use in an anterior lumbar surgical procedure. However, it is understood that the retractor system disclosed herein can be used in other surgical procedures, such as, but not limited to, cervical spine procedures and posterior facet surgery. The system **10** includes a frame **11** that is fixed to the operating table **T** by mounting components **12**. The mounting components **12** can be known mounting components for use in supporting surgical retractors or frames for retractors, such as the mounting components depicted in FIGS. **1** and **8**. The frame **11** can be a circular ring that is sized to avoid interference with the surgical site. The core of the present retractor system is the retractor **20** that retracts the tissue, blood vessels and organs, and defines a working channel for anterior access to the spine. The retractor **20** is supported on the frame **11** by brackets **13** with an engagement portion **14** that engage a leg of the retractor, as described in more detail herein. The brackets **13** are adjustably fastened to the frame **11** by clamping elements **15**.

(33) In one embodiment shown in FIGS. **12A-12B**, the retractor **20** includes a generally rectangular saddle-shaped tubular base **21** that defines a working channel **23** that is sized to receive surgical instruments and implants. The base **21** includes a lower rim **22** that is configured to generally conform to the vertebral bodies on either side of the intervertebral space, as shown in FIG. **11C**. In particular, the lower rim **22** can have a curvature that follows the curvature of the vertebral bodies **VB** at the point of surgical approach (i.e., anterior or posterior). It can be appreciated that the tubular base is sized to span the intervertebral space **I** so that the lower rim **22** can have adequate purchase on the adjacent vertebral bodies **VB**. Thus, the base **21** has a width **W** (FIG. **11C**) that is greater than the width of the intervertebral space **I**. As shown in FIG. **12A**, the lower rim **22** has a concave curvature that corresponds to the curvature of the vertebral bodies. The length **L** of the curved rim **22** is sufficient for the base **21** to form a solid foundation for the retractor when seated on the vertebral bodies **VB**. It can be appreciated that the curvature of the lower rim and the dimensions **W**, **L** of the generally rectangular base **21** are determined by the dimensions of the intervertebral space and vertebral bodies at the particular spinal level.

(34) Retractor **20** further includes a pair of projecting legs **24** projecting upward from the upper rim **25**. In one embodiment, the legs are integral with the base **21**. The legs define a fixation feature **32** that is engaged by the engagement portion **14** of the brackets **13** (FIG. **11A**), so that the retractor can be held in position by the fixed frame. Thus, the legs have a length from the upper rim that is sufficient for the fixation features to be outside the patient, and more particularly at a height of 7-10 cm above the patient's body. In a specific embodiment, the legs **24** have a length of 15-20 cm, while the base **21** has a height of 4-6 cm from the lower rim to upper rim.

(35) In one feature of the disclosure, the retractor **20** is integrally formed so that the legs **24** are one piece with the base **21**. The retractor **20** is formed of a radio-transparent or radio-lucent material that is sufficiently rigid to hold back the tissue, blood vessels and organs to maintain an adequate working channel for the surgeon. The material can be a thermoplastic, such as nylon, polycarbonates and polyketones, and may be a reinforced composite, such as a carbon-fiber thermoplastic. In another feature, the base **21** includes two portions **21a**, **21b**, with the upper portion **21a** being substantially rigid and the lower portion **21b** being generally compliant or flexible. The lower portion **21b** defines the lower rim **22** that contacts the vertebral bodies. The flexibility of the lower portion **21b** allows the portion to be compressed as the retractor **20** is pushed against the vertebral bodies (or pulled downward when used in a fixed system). The compliance of the lower portion allows it to conform to the local anatomy providing a firm base for seating the retractor.

(36) In one embodiment the two portions **21a**, **21b** can be formed of the same material using a manufacturing process that allows variation of the durometer or Shore hardness between the two

portions. In another approach the compliant portion **21b** can be adhered to the rigid portion **21a**, or can be molded onto the rigid portion in a two-stage molding process. In an alternative approach, the portions of the base can be produced in an additive or 3D printing process. The hardness and/or flexibility of the base **21** and the portions **21a**, **21b** can be controlled by the printing process to produce a base having various hardness or flexibility profiles. In one version, the two portions can have distinct hardness/flexibilities, whereas in another version the material properties can be varied at locations throughout the retractor body, meaning that more than two portions have different relative flexibilities/hardnesses. Additive or 3D printing also adds the ability to vary the hardness/flexibility in any direction of the retractor. For instance, the base **21** can have vertical sections of greater or lesser rigidity/flexibility or several horizontal regions of different rigidity/flexibility, all calibrated to retract particular tissue, such as nerves and blood vessels.

(37) Another embodiment of a retractor **20'** is shown in FIGS. **13A-D**. In this embodiment, a pair of upwardly projecting legs **24** is provided at opposite ends of the body **21**, and a second pair of legs **26** is provided at opposite sides of the body. As reflected in the view of FIG. **13B**, the body is elongated or generally rectangular so that the body can be positioned with its long axis **L** transverse to the axis of the spine. The body defines a working channel **23** that provides access to the surgical site. The width of the body between the legs **26** is greater than the width of a distracted intervertebral space so that the lower rim **22** can engage the adjacent vertebral bodies (FIG. **11C**). The legs **24**, **26** each include an angled tab **30** that defines the fixation features **32** that are connected to the engagement portion **14** of the brackets **13** used to connect the retractor to the frame, as described above. The fixation features **32** and engagement portions **14** are configured for removable engagement to allow the retractor to be positioned first within the surgical space and then connected to the frame **11**. In one embodiment, the fixation features **32** can be in the form of a keyhole or T-slot to receive a complementary configured connecting pin within the engagement portions **14** of the brackets **13**. In one embodiment, the fixation portion can be a slot defined by a large opening **32a** and a narrower opening **32b**. Thus, as shown in FIG. **14**, the engagement portion **14** can include edge panels **14a** that define a slot **14b** to receive a leg **24**, **26**. A post **14c** includes an enlarged head **14d** that can be inserted through the large opening **32a** and slid down into the narrower opening **32b** to fix the bracket **13** to the leg of the retractor **20'**. It can be appreciated that the orientation of the large opening relative to the narrower opening can be reversed from that shown in FIG. **13A** depending on the orientation of the bracket **13** engaged to the fixation feature **32**. If the bracket is oriented so that it exerts a downward force on the angled tab **30**, then the openings **32a**, **32b** can be arranged as shown in FIG. **13A**. On the other hand, if the bracket exerts a slightly upward force, the narrower opening **32b** should be above the large opening **32a** to retain the post **14c** within the fixation feature **32**.

(38) Returning to FIGS. **13A-13D**, the legs **24**, **26** have a length from the base **21** that places the fixation features **32** outside the patient's body and in alignment with the frame **11** for the fixed retractor system **10** (FIG. **11A**). The legs **26** can be angled outwardly or laterally from the sides of the base, as shown in FIG. **13D**, to help enlarge the working channel and improve access to the intervertebral space beneath the retractor **20'**. The legs **24** at the long ends of the retractor are generally vertically oriented, as shown in FIG. **13C**, to accommodate shims, as described herein. However, the legs **24** can also flare outwardly in the same manner as legs **26** in alternative embodiments.

(39) In one feature, the body **21** can define a plurality of bores **33** (FIGS. **13A**, **13B**) extending through the side walls of the body **21** and opening adjacent the rim **22**. The bores can receive anchor pins **P** or screws that can be driven into the adjacent vertebral bodies to solidly anchor the retractor (FIG. **11B**), particularly when the retractor is used as part of a self-supporting system.

(40) It is understood that the retractor **20'** can be formed of the same material as the retractor **20** discussed above. Moreover, the base **21** of the retractor **20'** can be configured like the base shown in FIGS. **12A-12B** to include the rigid upper portion **21a** and the compliant lower portion **21b**.

(41) Another retractor **40** is shown in FIGS. 15A-C that is akin to a conventional tubular retractor. In particular, the retractor **40** includes a frusto-conical tubular base **41** having a generally elliptical or elongated cross-section similar to the retractor **20'** described above. Unlike the prior retractors **20, 20'**, the body **41** of the retractor **40** spans the space between the vertebral bodies and the fixation frame **11** (FIG. 11A). The upper rim **41a** of the body merges into a shallow disc **43** which defines the fixation features **44** for engaging the fixation frame to the retractor **40**. The disc **43**, and particularly the fixation features **44**, define a diameter that is substantially the same as the diameter of the fixation frame **11** so that the frame can be engaged directly to the disc without the need for the brackets **13**. The lower portion **46** of the base **41** is open and defines the lower rim **41b** for passage of tools, instruments and implants into the surgical site. As with the other self-supporting retractors, the lower rim **41b** is configured to engage the surface of adjacent vertebral bodies. The lower portion **46** can thus be compliant or flexible, like the lower portion **21b** of the retractor **20** described above. The lower portion **46** need not be symmetrical but may instead adopt different contours to comply with the local anatomy. For instance, as shown in FIG. 15A, the lower portion can define a compliant or flexible lip **47** projecting from one side of the lower rim **41a** that can be used to navigate the retractor past a vessel or nerve.

(42) The body **41** can define an attachment feature **48** for connecting to additional instruments or tool. Similar attachment features **48** can be provided on the outer surface of the body **21** of the retractor **20, 20'**, most particularly along the width of the retractor. The attachment points are configured to engage a hand-held retractor, such as the retractor **49** shown in FIG. 16. The attachment feature **48** can be configured as a key-slot, as shown in FIG. 15B, and the hand-held retractor **49** can incorporate a post **49e** adapted to be received in the attachment point, similar to the post **14c** (FIG. 14). The hand-held retractor includes a generally horizontal portion **49a** that can be a rod terminating in a knob **49b** configured for manual manipulation of the retractor from outside the patient's body. The horizontal portion **49a** merges into a vertical portion **49c** that can be a plate sized to extend along one of the legs of the retractor **20, 20'**. The end portion **49d** of the vertical portion **49c** can be relatively more flexible than the remainder of the hand-held retractor **49** to reduce the risk of trauma to surrounding tissue when the retractor **49** is used to introduce a self-supporting retractor **20, 20'**. The hand-held retractor can be used by the surgeon to navigate the retractor **20/20'/40** into position.

(43) In a further feature of the retraction system disclosed herein, the retractor, particularly retractor **20'**, can be configured to engage and support shims. As described above, a shim can act as a distractor by contacting the endplates of the adjacent vertebral bodies and holding a disc space open. The shim can also help anchor the retractor in position as the adjacent vertebral bodies bear against the shim, holding it, and therefore the retractor attached to the shim, in place. In one embodiment, the end legs **24** of the retractor define an elongated recess **35** extending at least from the upper rim of the base **21** to the lower rim **22** and aligned with a corresponding leg (FIGS. 13B, 17). The recess **35** is thus in the form of a vertical channel that slidably receives a shim, such as the shim **50**, shown in detail in FIGS. 18A-E. The shim **50** includes an elongated plate portion **52** that is sized to slide into the recess **35** on either leg **24**, as shown in FIG. 17. The lower portion **51** of the shim **50** is configured as a wedge or distractor. The shim is thus introduced into the retractor **20'** when the rim **22** is seated on the adjacent vertebral bodies, with the lower portion **51** projecting below the rim **22**, into the intervertebral space and in contact with the endplates of the adjacent vertebral bodies. The lower portion **51** not only anchors the retractor to the vertebral space but also helps distract and maintain the intervertebral space between the adjacent vertebral bodies. It is understood that the width of the lower portion **51** must be calibrated to the particular vertebral level and desired distance between the adjacent vertebral bodies. The recess **35** in the retractor **20'** is thus wider than the maximum width distractor.

(44) The distractor **50** is held in place by a resilient latch **53** and post **54** that can flex inward from the locking position shown in FIG. 18C. The latch flexes inward as the shim **50** is introduced into

the recess **35** of the retractor **20'** until the post **54** clicks into an indentation **55** (FIG. **13A**) defined in the retractor. Other latching or locking mechanisms are contemplated to solidly attach a shim **50** to each leg **24**.

(45) The present disclosure contemplates a self-supporting embodiment of the retractor **20'**, as shown in FIG. **19**. In this embodiment, the retractor can be at least partially anchored to the vertebral bodies by pins or screws through bores **33**, as described above, and then the legs **24**, **26** can be maintained in a proper position by the frame assembly **60**. The tension in the soft tissue surrounding the retractor can also help hold the retractor in position within the surgical site as the tissue bears against the legs **24/26** of the retractor. The frame assembly **60** includes a ring **61** that can be configured to encircle the surgical site, such as with a rectangular, trapezoidal, oval or circular shape. It is contemplated that the perimeter dimension of the ring **61** is calibrated to avoid impeding the surgeon's access to the working channel **23** within the retractor **20'**. The frame is configured to engage an adjustable connector component **65**, as described below. The frame can include projections **62** that are configured to be manually grasped to facilitate manipulation of the frame during installation of the self-supporting retractor. In addition, or alternatively, the projections **62** can serve as fiducials for imaging purposes and/or for use in connection with a surgical navigation system. In this instance, the projections **62** are at least coated with a material for imaging the projection, whether by an optical detector or by X-ray. It is further contemplated that similar fiducials may be incorporated into the retractor **20/20'/40** and/or shims **50** engaged to the retractor, provided that the fiducials are offset from the working space **23** at the surgical site. In one embodiment, a fiducial similar to the projection **62** can extend from the outer face of one or more of the legs **24/26**, or a radio-opaque marker can be embedded within any part of the retractor or shims. The retractors disclosed herein are effectively anchored to the spine and remain anchored throughout the surgical procedure. Fiducials incorporated into the retractor and/or shims can be used to prevent errors in registration between the image used in the surgical navigation and/or to help register the images to the anatomy.

(46) The adjustable connector component **65**, shown in more detail in FIGS. **20-26**, includes a barrel mechanism **66** that receives a flexible strap **68** therethrough, as shown particularly in FIGS. **20-21**. The strap **68**, shown in FIGS. **22A-22D**, includes an elongated body **69** with a series of ridges **70** on one side of the body, similar to the well-known "zip tic". The strap terminates in an engagement portion **71** that includes a retainer bar **72** and an engagement groove **73**. The retainer bar **72** is configured to be introduced in one orientation into the slot **32** (FIG. **12A**) of the retractor legs **24**, **26**, and then rotated 90° to an orientation in which the retainer bar cannot pass through the slot. The tab **30** is engaged within the groove **73** to effectively clamp the engagement portion **71** onto the tab. In an alternative embodiment, the retainer bar **72** can be circular, in which case the slot **32** can have the configuration shown in FIG. **13A**. In either configuration, the slot **32** and retainer bar **72** are configured to allow the retainer bar to enter the slot in one position and then latch onto the retractor leg in another position.

(47) The barrel mechanism **66** includes a body **80**, a barrel **81** and end cap **82**, as depicted in FIG. **21A**. As shown in the detail views of FIGS. **23A-D**, the body **80** defines a slot **84** for receiving the ring **61**, with a recess **86** formed inside the slot to retain the body on the ring when the components are tightened, as described herein. When the body **80** is mounted on the ring and the strap **68** is tightened, the ring is firmly seated in the recess **86** to hold the barrel mechanism in position. The body **80** further defines a channel **87** through which the strap is threaded, as depicted in FIG. **21A**. A sliding surface **88** extends along the bottom of the channel **87** to provide support for the strap as it is engaged by the barrel **81**. In one embodiment, the upper surface of the channel can be provided with features for engaging the ridges **70** of the strap **68** to control, but not prevent, longitudinal movement of the strap.

(48) The body **80** also includes resilient mounting features **89** that are adapted to support the barrel **81** on the body. The barrel is provided with an external gripping feature **92** that allows the barrel to

be manually gripped and rotated by the surgeon (FIGS. 25A, C). The inner circumferential surface 93 of the barrel includes surface features 94 defining a control portion configured to engage the ridges 70 of the strap to control longitudinal movement of the strap and with sufficient strength to maintain the strap in tension when the strap is tightened between the ring 61 and the retractor legs 24, 26. The control portion surface features 94 are essentially ratchets or ridges that engage the ridges 70 of the strap such that all longitudinal movement of the strap is prevented, thereby locking the connector component 65 to the ring 61. The resilient mounting features 89 are configured to allow the barrel 81 to be moved upward relative to the sliding surface 88 so that the ridges of the surface features 94 disengage from the ridges 70 of the strap 68. The mounting features 89 are thus configured as V-shaped legs projecting laterally outward from the body 80. The legs can be bent or deflected upward by moving the barrel 81 upward, but the legs still maintain pressure on the barrel from inside the barrel to hold the barrel in position on the body 81. The barrel can be moved upward until the inner circumferential surface 93 contacts the flanges 81a that are diametrically opposite the sliding surface 88 of the body. The resilience or “springiness” of the mounting features 89 push the barrel back into engagement with the strap 68 when manual pressure is released from the barrel.

(49) In another embodiment, the barrel is supported by the mounting features 89 for rotation along the longitudinal axis of the barrel mechanism. In this embodiment, the barrel 81 is essentially a rotating nut that is mounted on the mounting features 89. In this embodiment, the interior of the barrel also includes a free portion 95 between the surface features 94 that does include the surface features and therefore does not engage the ridges of the strap so that the strap can move longitudinally freely through the channel 87. The barrel 81 can be rotated to a first position in which the free portion 95 is aligned with the ridges of the strap 68 so that the strap can be freely tightened. In this position, the surgeon can pull the elongated body 69 of the strap to place the strap in an initial tension. The surgeon can then rotate the barrel 80 until the control portion 84 is juxtaposed with the ridges 70 of the strap. The interface between the control portion 84 and the ridges 70 can be configured to pull the strap longitudinally with small rotations of the barrel 81 relative to the body 80, such as providing the control portions with a slight helical curvature, akin to internal threads. This configuration can allow fine tuning the tension of the assembled retractor system. In this embodiment, the barrel 81 can include external indicia to identify whether the barrel is in the first or second positions—i.e., whether the control portion 84 or the free portion 85 is aligned with the ridges 70 of the strap.

(50) The assembly of the barrel mechanism 66 is completed by the end cap 82 shown in FIGS. 26A-26D. The end cap 82 includes an end plate 98 from which a mating body 96 projects, as shown in FIG. 26A. The mating body is configured to be received within a bore 90 defined in the body 80 of the barrel mechanism (FIG. 23A). A peg 97 engages an opening 91 (visible through the channel 86, as seen in FIG. 23B) in the body 80 to fix the end cap 82 to the body 80, with the barrel 81 sandwiched in between. The end plate defines a channel 99 that aligns with the channel 87 in the body 80 to receive the strap 68. It can be appreciated that the connector component 65 can be provided fully assembled to the surgeon, as depicted in FIG. 21A, with the barrel in the rotational position to hold the strap in place within the barrel mechanism.

(51) As with the retractor, the entirety of the clamping component 65 is formed of a radio-transparent or radio-lucent material, such as plastic. The material for the ring 61 must be sufficiently rigid to provide the necessary support when the straps 68 are all placed in tension, but can still be formed of a translucent material.

(52) In an alternative embodiment, a self-supporting system 60' is shown in FIG. 27 in which the ring 61' is directly mounted to a retractor, such as retractor 20'. In particular, the ring 61' is attached to the angled tab 30 at the end of each leg of the retractor 20'. The ring 61' can include downward posts (not shown) configured to engage the fixation features 32, which can be a key-slot configuration. In this embodiment, the legs 24, 26 of the retractor can be curved outwardly from

the base of the retractor, as depicted in FIG. 27, so that the fixation features are aligned with the ring **61'**. This feature allows the ring **61'** to define essentially the same circumference as the ring **62** in FIG. 19. In this embodiment, the posts can have a circular head and the fixation features **32** can be configured as show in FIG. 13A to receive the circular head in the larger portion of the fixation feature.

(53) It is further contemplated that lighting elements are incorporated into the retraction system disclosed herein. The lighting elements can be mounted to the ring **11** of the fixed system or the ring **61** of the self-supporting system, or can be integrated into the legs **24**, **26** of the retractor **20** or into the shims **50**. Alternatively, or in addition to the ring-mounted lighting, lighting can be integrated into the retractor itself. In one embodiment, a light source **100** is integrated into the legs **24** of the retractor **20**, **20'** as shown in FIG. 28A, or into the shims **50**, as shown in FIG. 28B. The portion of the legs or shim in which the light source is embedded can be formed of a translucent material, such as a plastic, that allows light to pass through the body and that diffuses the light to illuminate a surgical site. Light can be conveyed to the light source **100** by a light transmission component **101**, which can be a light pipe or fiber optic cable as is known in the art. This approach does not increase the overall bulk or envelop of the retractor and provides significant lighting for the surgeon.

(54) Alternatively, or in addition, a directional light **102** can be incorporated into a shim **50**, as shown in FIG. 29. The light **102** can be mounted within a recess **104** in the shim and connected to light pipe or fiber optic cable **103** in a conventional manner. In this embodiment, the light **102** shines directly onto the surgical site, rather than being diffused as in the prior embodiments.

(55) It is contemplated that rather than constitute a light source, the components **100** or **102** can be a visualization device, such as a camera or optical detector, that permits visualization of the surgical site within the retractor. Thus, in one embodiment, two such components **100/102** are provided, with one component constituting a light source and the other component constituting a visualization device. The image detected by the visualization device can be transmitted along the transmission component **101** to a receiver (not shown) that can be used to display the image for the surgeon's benefit during the procedure.

(56) An alternative retractor **110**, shown in FIGS. 30A-B, is similar to the retractors **20**, **20'**, but modified to permit adjustment of two legs of the retractor. The retractor **110** includes a base **111** with legs **112** extending from the sides of the retractor, similar to the legs **26** of the retractor **20'** (FIG. 13D). However, rather than the fixed legs **24**, the legs **115** are pivotable within a channel **117** defined in the base **111** of the retractor. The interface between the pivot end **116** of the leg **115** and the channel **117** allows the leg to be moved from one side of the retractor (FIG. 30a) to the other (FIG. 30B). This feature allows the surgeon to tailor the location of the legs **115** based on the anatomy being retracted and/or on the angle of approach that the surgeon is taking to access the spine. It is contemplated that the pivot interface between end **116** and channel **117** can allow the leg **115** to be pivoted at any time during the surgery and as needed by the surgeon to access the site. The interface between the pivot end **116** and the channel **117** can be a close running fit or a friction fit that permits limited movement of the leg **115** but provides a sufficient frictional engagement between the components to hold the position of the leg once adjusted by the surgeon. Alternatively, a separate fixation mechanism can be incorporated into the pivot interface to lock the leg **115** into a particular orientation.

(57) The present disclosure should be considered as illustrative and not restrictive in character. It is understood that only certain embodiments have been presented and that all changes, modifications and further applications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A surgical retractor for retracting soft tissue and anatomy at a spinal surgical site, comprising: a tubular base defining an upper rim and a lower rim and a working channel between the upper and lower rims, the base and lower rim sized and configured to be seated on adjacent vertebral bodies spanning an intervertebral space, wherein said tubular base includes; an upper portion defining said upper rim; and a lower portion defining said lower rim and connected to said upper portion, wherein said lower portion is relatively more flexible than said upper portion; and at least two elongated legs, each projecting from said upper rim and each including a fixation feature at a free end thereof, each of said at least two legs having a length sufficient for the fixation feature to be outside the body of a patient when the lower rim is seated on adjacent vertebral bodies of the patient, wherein said base and said at least two legs are formed of a radio-transparent or radio-lucent material.
2. The surgical retractor of claim 1, wherein said base and at least two legs are integrally formed.
3. The surgical retractor of claim 1, wherein said upper portion and said lower portion of said tubular base are integrally formed.
4. The surgical retractor of claim 1, wherein said lower rim has a length and a width, and defines a concave curvature along said length sized to correspond to the curvature of a vertebral body.
5. The surgical retractor of claim 1, wherein said base defines at least one bore extending from said upper rim to said lower rim, said at least one bore sized to receive an anchoring pin for anchoring the retractor to a vertebral body.
6. The surgical retractor of claim 1, wherein each of said two legs includes an end portion defining said fixation feature, said end portion angled outward relative to a portion of each of said two legs between said end portion and said base.
7. The surgical retractor of claim 1, further comprising a light source mounted to at least one of said at least two legs, said light source arranged to direct light into said working channel.
8. The surgical retractor of claim 7, wherein said light source is embedded within said at least one of said at least two legs, and said at least one of said at least two legs is at least partially formed of an optically translucent material.
9. The surgical retractor of claim 7, further comprising a visualization component mounted to at least one of said at least two legs, said visualization component configured to visualize the surgical site through the working channel.
10. The surgical retractor of claim 1, further comprising: said tubular base defining at least one elongated recess extending from said upper rim to said lower rim and aligned with a corresponding leg of said retractor; and an elongated shim having an elongated plate portion slidably received within said at least one elongated recess and an integral lower portion extending below said lower rim when said elongated plate portion is seated within said at least one recess, said lower portion having a width sized to contact the endplates of the adjacent vertebral bodies when the surgical retractor is seated on the vertebral bodies.
11. The surgical retractor of claim 10, further comprising a resilient latch for removable engagement between said tubular base and said elongated plate portion of said shim.
12. A surgical retractor system comprising: a surgical retractor according to claim 1; a frame; and at least two brackets, each engageable to said fixation feature and each mountable between said frame and said fixation feature of a corresponding one of said at least two legs.
13. The surgical retractor system of claim 12, further comprising a connector component for engaging each of said at least two brackets to said frame.
14. The surgical retractor system of claim 13, wherein: each of said at least two brackets is an elongated strap having; an engagement portion at one end configured to engage said fixation feature of said corresponding one of said at least two legs; and a series of ridges at an opposite end thereof, and said connector component is configured to selectively engage said series of ridges to fix said elongated strap relative to said connector component and to selectively disengage said

series of ridges to permit movement of said elongated strap relative to said connector component.

15. The surgical retractor system of claim 14, wherein said connector component includes: a body defining a channel through which said elongated strap extends; and a barrel rotatably mounted to said body in alignment with said series of ridges of said elongated strap, said barrel defining internal surface features configured to engage said series of ridges when said barrel is in a first position and an internal free portion that does not engage said series of ridges when said barrel is in a second position.

16. The surgical retractor system of claim 12, wherein said frame is a rigid ring and includes a projection extending from said rigid ring, said projection configured to be manually grasped to manipulate said rigid ring.
